

UNIVERSITY OF OSLO

BedreFlyt Artifact: A Self-Adaptive Digital
Twin Software System for Hospital
Resource Allocation

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ACSOS 2026

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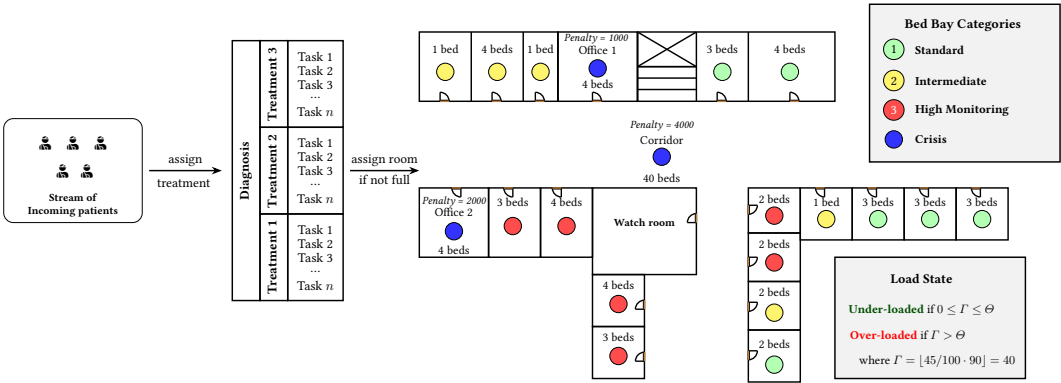
September 10, 2026



Contents

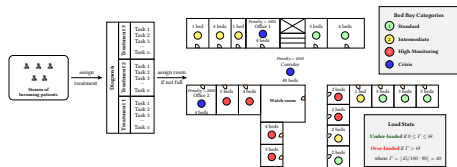
- ① Rationale
- ② Digital Twin Concepts
- ③ Digital Twin Architecture

What do we need?



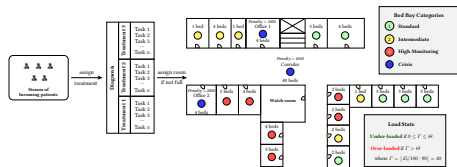
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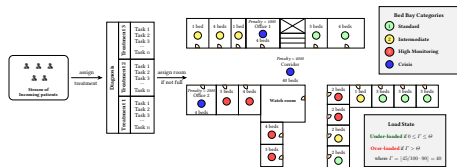
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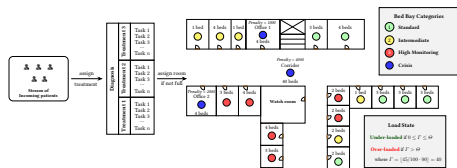
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- Depends on several factors

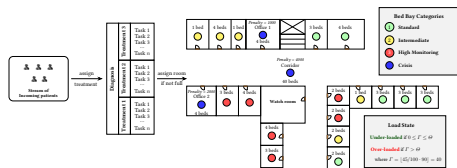


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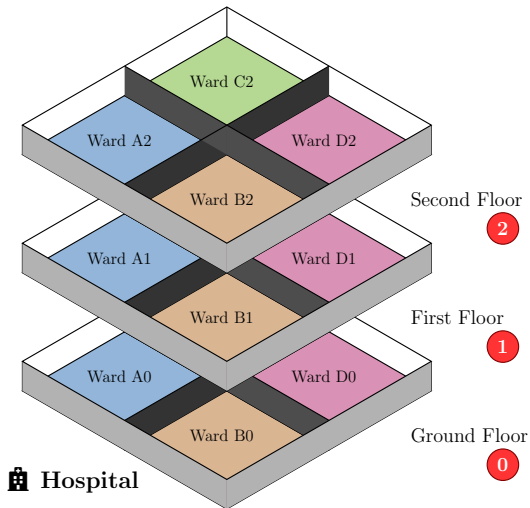


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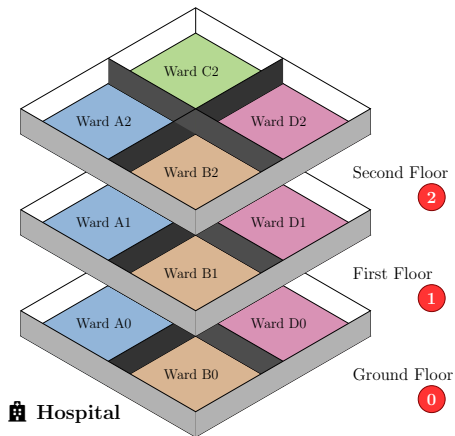


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- Admission is often handled manually
- Depends on several factors
- Static architectures struggle when the underlying structure changes
- We want a tool that can help us in the allocation process

What do we need? (cont)

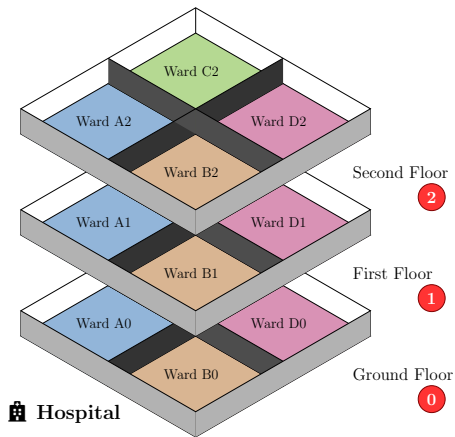


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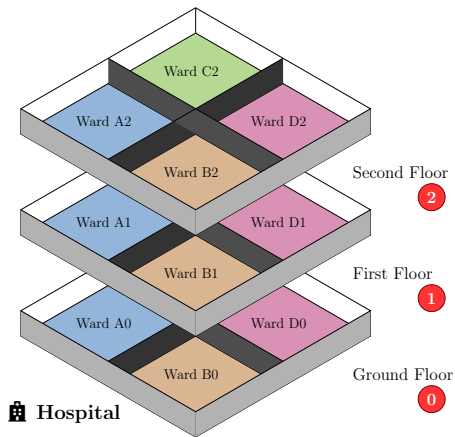
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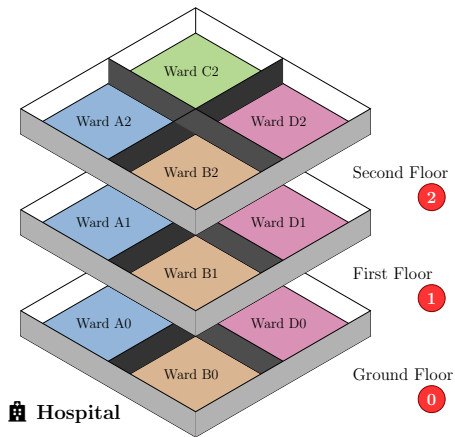
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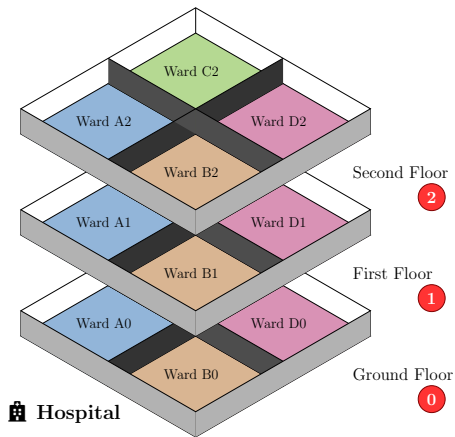
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- If not enough space is free they might be moved
- This requires knowledge of the layout and strategies for interaction
- We want to place the patient as close as possible to the intended ward

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What is a Digital Twin

Digital twins are, as defined by NASEM

NASEM Definition of DT (2024)

A digital twin is a set of virtual information constructs that mimics the structure, context, and behaviour of a natural, engineered, or **social system** (or system-of-systems), is dynamically updated with data from its physical twin, has a predictive capability, and **informs decisions** that realise value. The bidirectional interaction between the virtual and the physical is central to the digital twin^a.

^aNational Academies of Sciences, Engineering, and Medicine (NASEM) (2024): Foundational Research Gaps and Future Directions for Digital Twins. The National Academies Press

Twinning in SMOL

Programming support for twins with a behavioural and a structural layer
smolang.org



Twinning in SMOL

Programming support for twins with a behavioural and a structural layer

SMOL: Semantic Model Object Language

- Smalll OO programming system
- Ontology reasoners allow querying the KB
- Used to create the digital model

smolang.org



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behavioural twins in SMOL

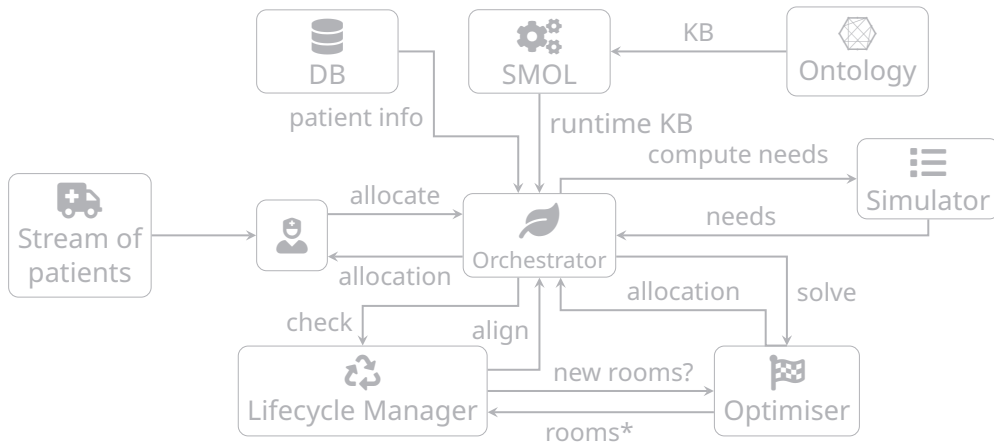
- SMOL can encapsulate (simulation) models based on the FMI standard
- Can automatically adapt the object states in the KB at runtime through semantic lifting



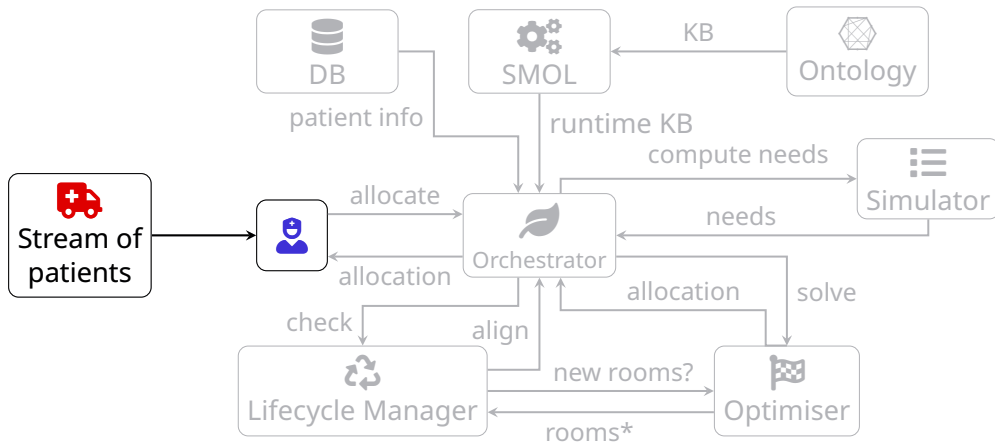
Contents

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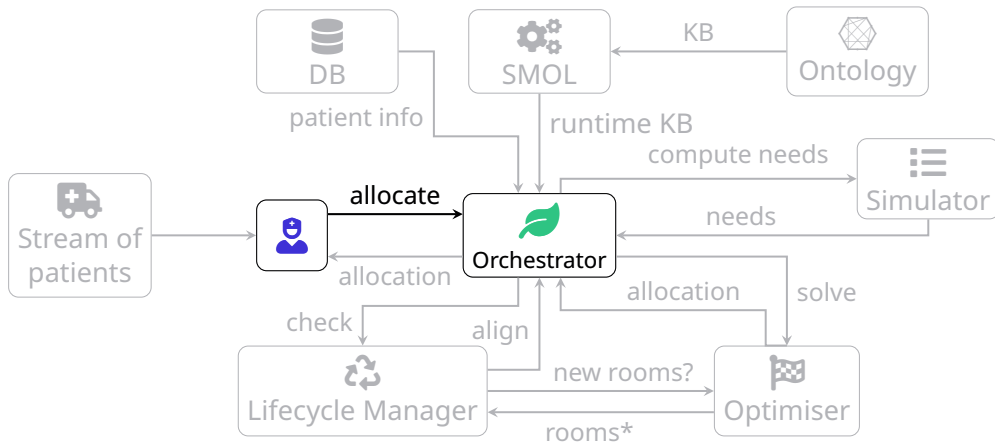
Allocation of patients



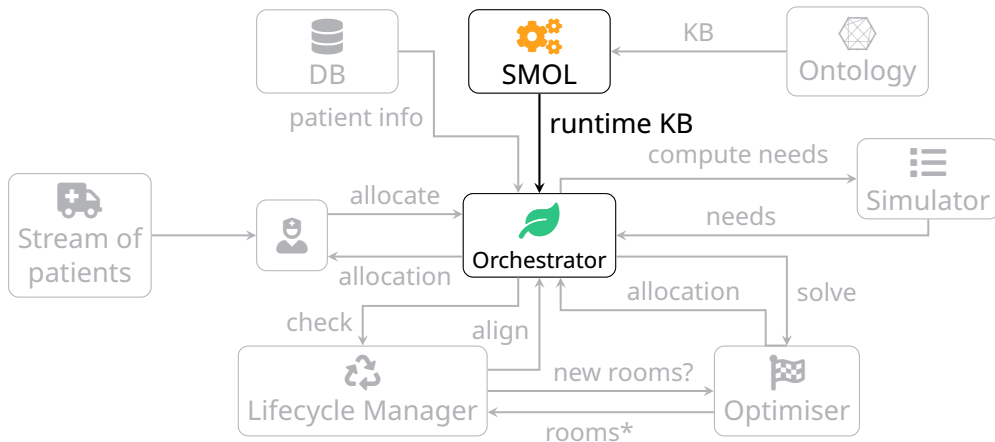
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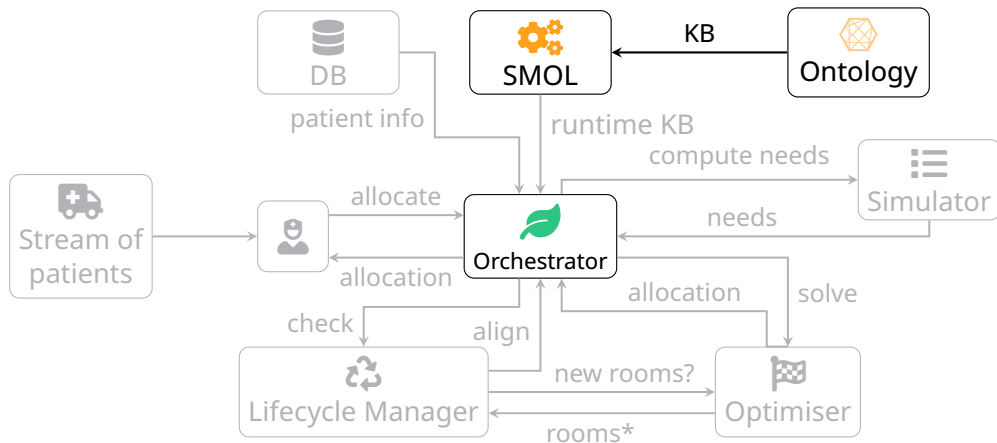
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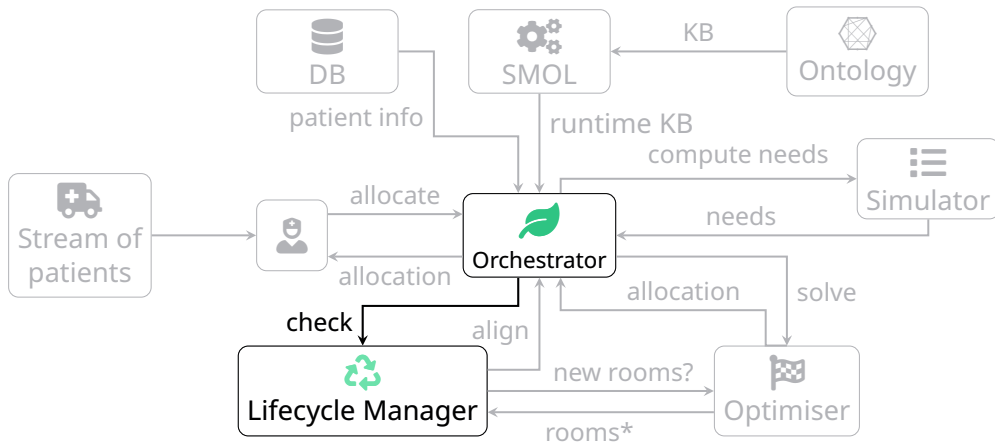
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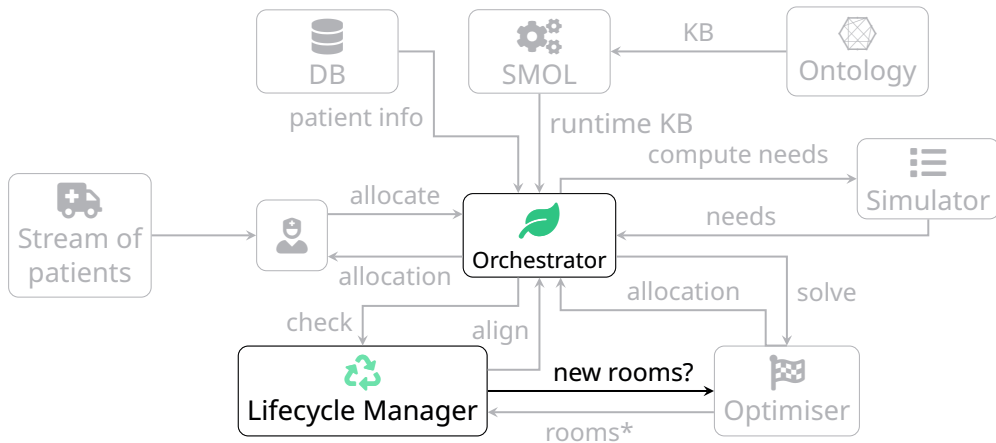
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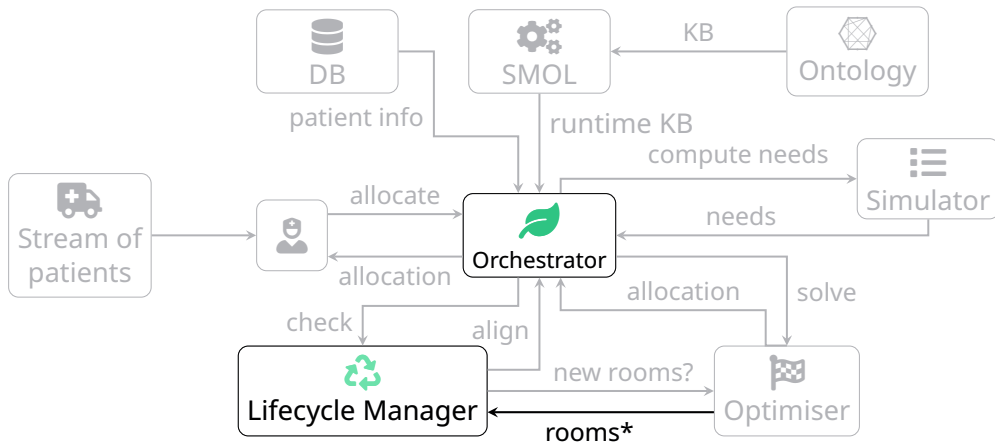
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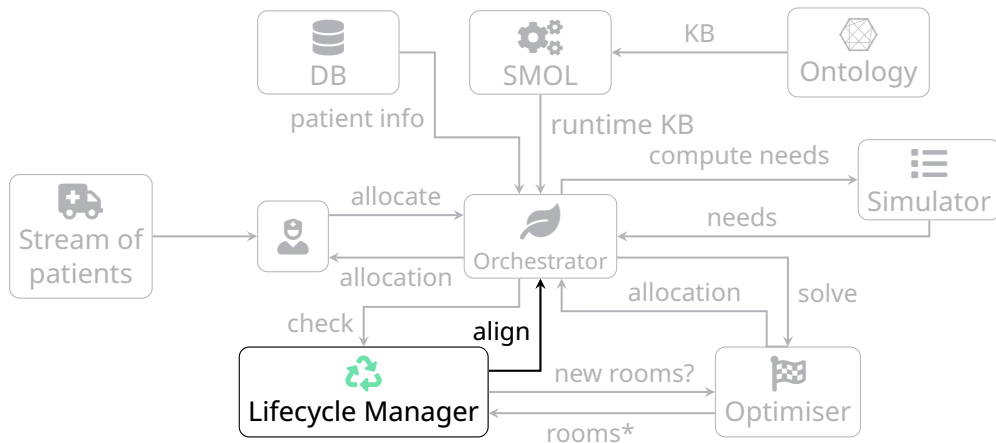
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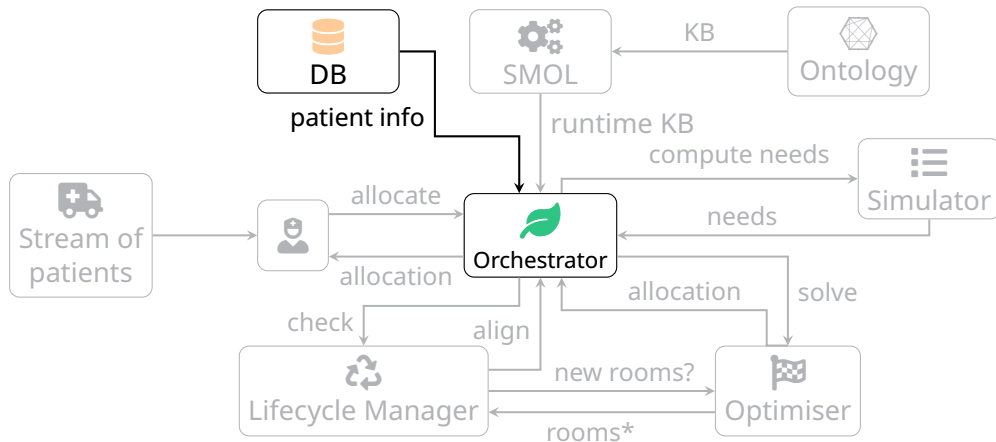
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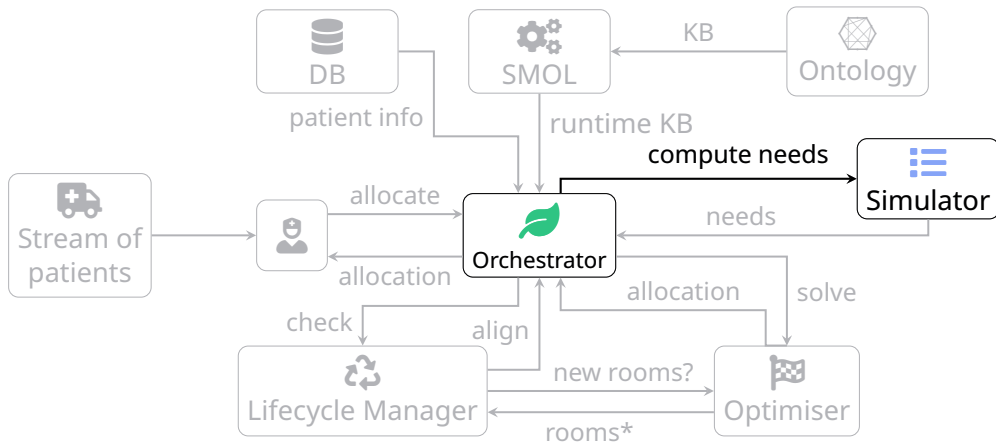
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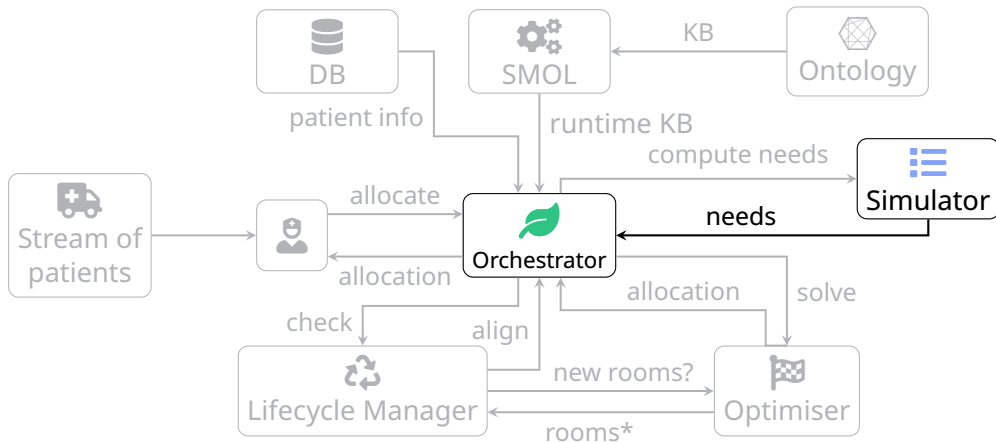
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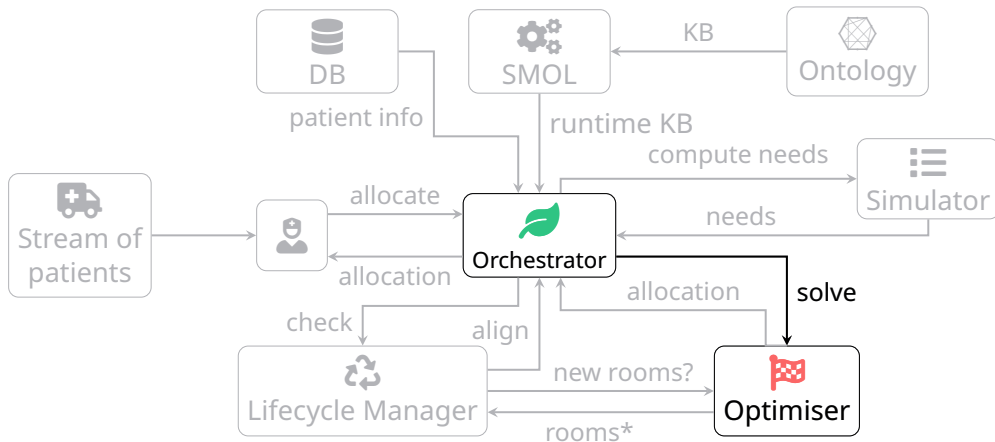
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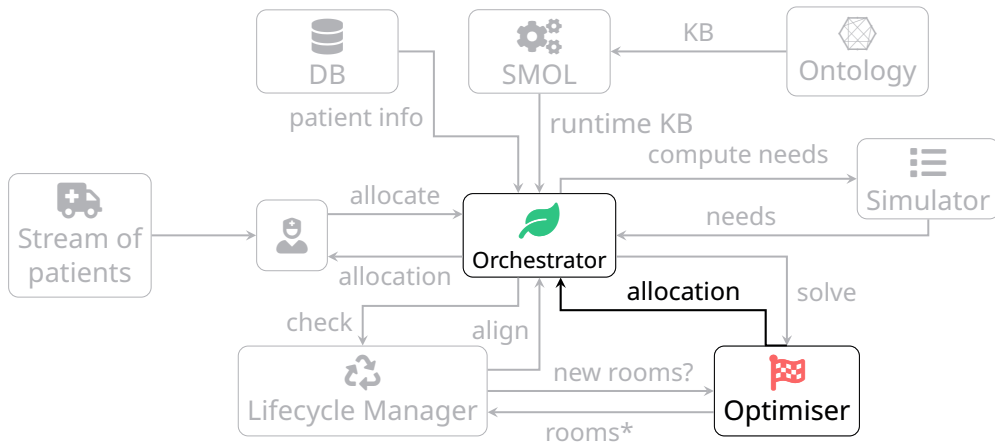
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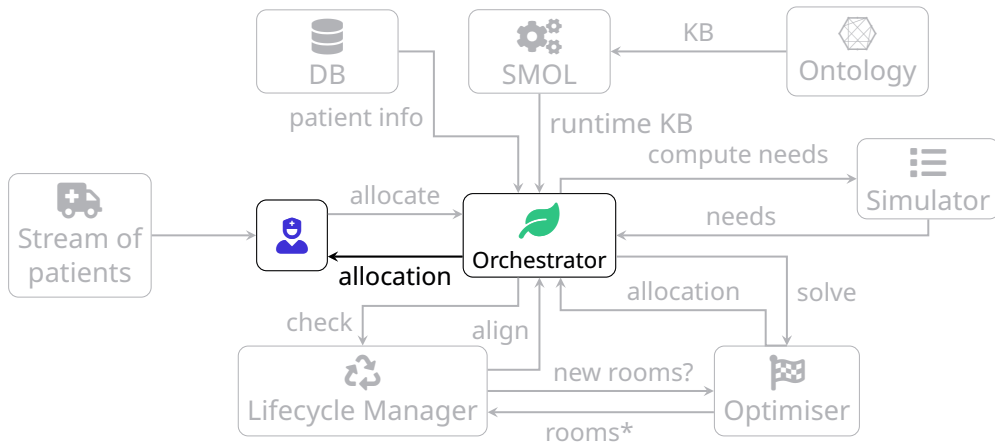
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Human-in-the-loop in the allocation

- The DT proposes the *allocation of patients*

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- The DT proposes the *allocation of patients*
- The human-in-the-loop remains the final authority for adaptation actions
- Approved actions are reflected back via semantic lifting to ensure the model is aligned
- This enables supervised autonomy: controlled self-adaptation instead of static decision support

Run the DT

The screenshot shows the BedreFlyt DT web application interface. On the left is a dark vertical sidebar with several icons; the icon representing a document or report is highlighted with a red rectangle. The main content area has a dark header bar with the 'BedreFlyt DT' logo and a user profile icon. Below the header, there are three tabs: 'Allocation' (selected), 'Simulation', and 'Current Allocation'. The 'Allocation' tab contains a section titled 'Patients to allocate' with an 'Allocate' button. Below this is a 'Ward' dropdown menu currently set to 'Ortopedi - OSL-OUS'. To the right of the dropdown is a checkbox for 'Self-adaptive allocation' which is checked. Below these elements is a green button labeled '+ Add Patient'. At the bottom of the 'Allocation' tab is a 'Results' section which currently displays the message: 'No results available. Run a simulation to see results.'

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- It is also possible to use the Event Stream Generator as a CLI tool simulate stream of patients
- The modularity of the Twin allows the system to change the components as long as the input and output remains consistent
- The DT is being kept updated on GitHub
(<https://github.com/BedreFlyt/bedreflyt>)

Future directions

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- **Live data integration** connect to ward systems rather than simulator surrogate
- **Broader scope** multi-ward, staff constraints, other domains
- **Live artifact** make the GitHub repo a live artifact where the new components are merged and the contributors can work on

Thanks for the attention

Questions?

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